

JOUF UNIVERSITY

Medication Tracking Application For Elderly Users

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College of Computer and Information Sciences | 2025

The Adherence Crisis

Medication adherence is a critical global healthcare challenge, particularly among elderly populations managing multiple conditions.

 **50% Non-Adherence:** Approximately half of all elderly patients fail to take medications as prescribed.

 **Severe Health Outcomes:** Associated with 125,000 preventable deaths annually.

 **Hospitalizations:** Accounts for at least 10% of all hospital admissions.

 **Economic Burden:** Estimated at \$100-\$300 billion annually in the US alone.

50%
of elderly patients miss doses

Challenges for the Elderly



Polypharmacy

Prevalent among older adults (44% men, 57% women over 65). Taking 5+ medications daily causes massive scheduling confusion and dosage errors.



Cognitive Decline

Age-related forgetfulness, visual impairments, and complex UX in existing apps make it difficult to read labels and log doses.



Caregiver Gap

Family members lack real-time visibility into medication adherence, preventing timely interventions when doses are missed.

Review of Current Solutions

An evaluation of existing medication management applications revealed significant shortcomings:



Apple Health

Excellent tracking features, but strictly exclusive to iOS devices, alienating Android/Windows users.



Medisafe

Feature-rich, but hides essential caregiver features behind premium paywalls.



MyTherapy

Good interface, but crucially lacks advanced drug interaction warnings.



EveryDose & Dosecast

Good reminder systems, but manual entry is tedious and visual designs can be outdated.

Competitor Analysis

App	Cross-Platform	Privacy	Open-Source	Customization	Free	Internationalization
Apple Health	No	Partially	No	No	yes	Yes
Medisafe	Yes	Partially	Partially	Yes	Yes	Yes
MyTherapy	Yes	Partially	No	Partially	Yes	Yes
EveryDose	Yes	Partially	No	Yes	Partially	Yes
Round Health	No	Partially	No	No	Yes	No
Dosecast	Yes	Partially	No	No	Partially	No
Our Solution	Yes(Web, Win, Mac, Linux)	Yes	Yes	Yes	Yes	Yes

Our Proposed Solution

A specialized medication tracking application tailored for elderly users and their caregivers.



Dual Deployment

Distributed as standalone native desktop applications and an accessible web application.



Privacy

Local SQLite storage. No mandatory cloud synchronization or third-party data sharing.



Open-Source Transparency

Publicly auditable codebase ensures maximum security and community improvement.



Zero Financial Barrier

Completely free to use with no premium tiers.

Cross-Platform & Open Source



Cross-Platform

Develop once, deploy everywhere. Access from Windows, macOS, and Linux without compromise.

Lightweight Builds:

- Linux: ~11 MB
- Windows: ~14 MB
- macOS: ~4 MB



Open-Source

The entire source code is publicly accessible and auditable, fostering community trust and security.

Activity Metrics:

- 110 Commits
- 33 Issues Managed
- 34 Pull Requests Merged



Security Auditing

Invites global scrutiny to verify data protection standards on every pull request.

Key Advantage:

"Removed an entire class of 'it works on my machine' errors through automated CI/CD."

Our solution utilizes the **Wails runtime**, delivering binaries that are an **order of magnitude smaller** than equivalent Electron-based applications. This ensures high performance even on limited hardware, maintaining a consistent experience from v0.0.1 to the current v0.5.0 without manual compilation.

Primary Stakeholders



Patients (Elderly Users)

Interact with simplified large-font reminders, confirm doses, and track adherence history. Beneficiaries of customizable alerts.



Caregivers

Organize schedules, manage prescription refills, and receive missed dose alerts to intervene promptly and monitor compliance.



System Administrators

Maintain technical infrastructure, database security, perform updates, and ensure data protection standards are upheld.

System Analysis: Use Cases

The system defines distinct workflows to separate patient simplicity from caregiver complexity:



Patient

Receive reminders, confirm doses, log vital signs, and generate adherence reports.



Caregiver

Manage profiles, add medications, set flexible schedules, and monitor adherence remotely.



Automated

Independent background calculations for adherence rates, missed dose alerts, and automated refill reminders.

System Interactions (Sequence)

Sequence interactions define the strict temporal flow of the application's core logic:



Authentication

Verifies credentials against an encrypted SQLite database, generating secure session tokens.



Add Medication

Queries internal drug database, stores records securely, and queues precise notification alerts.



Dose Confirmation

Immutable updates to database logs (Taken/Skipped), cancels pending reminders, and updates inventory.



Missed Dose Alert

Parallel notifications sent to patient and caregiver interfaces if 30 minutes pass without dose confirmation.

Database Schema Details

Data integrity and full privacy are ensured through standard local tables without cloud reliance:



Users, Patients, Caregivers

Manages roles, secure bcrypt passwords, and strict relational associations.



Schedules and Medications

Stores start/end dates, hourly intervals, custom instructions, and structured drug types.



Logs and Notifications

Precise logging of 'TakenAt' timestamps and tracking of unread alert states.



Measurements and Reports

Tracks essential vital signs and paths for generated PDF documentation.

UI Design: Authentication

 pills



Welcome

Sign in to access your medical system

Email

Password

☒ Remember me

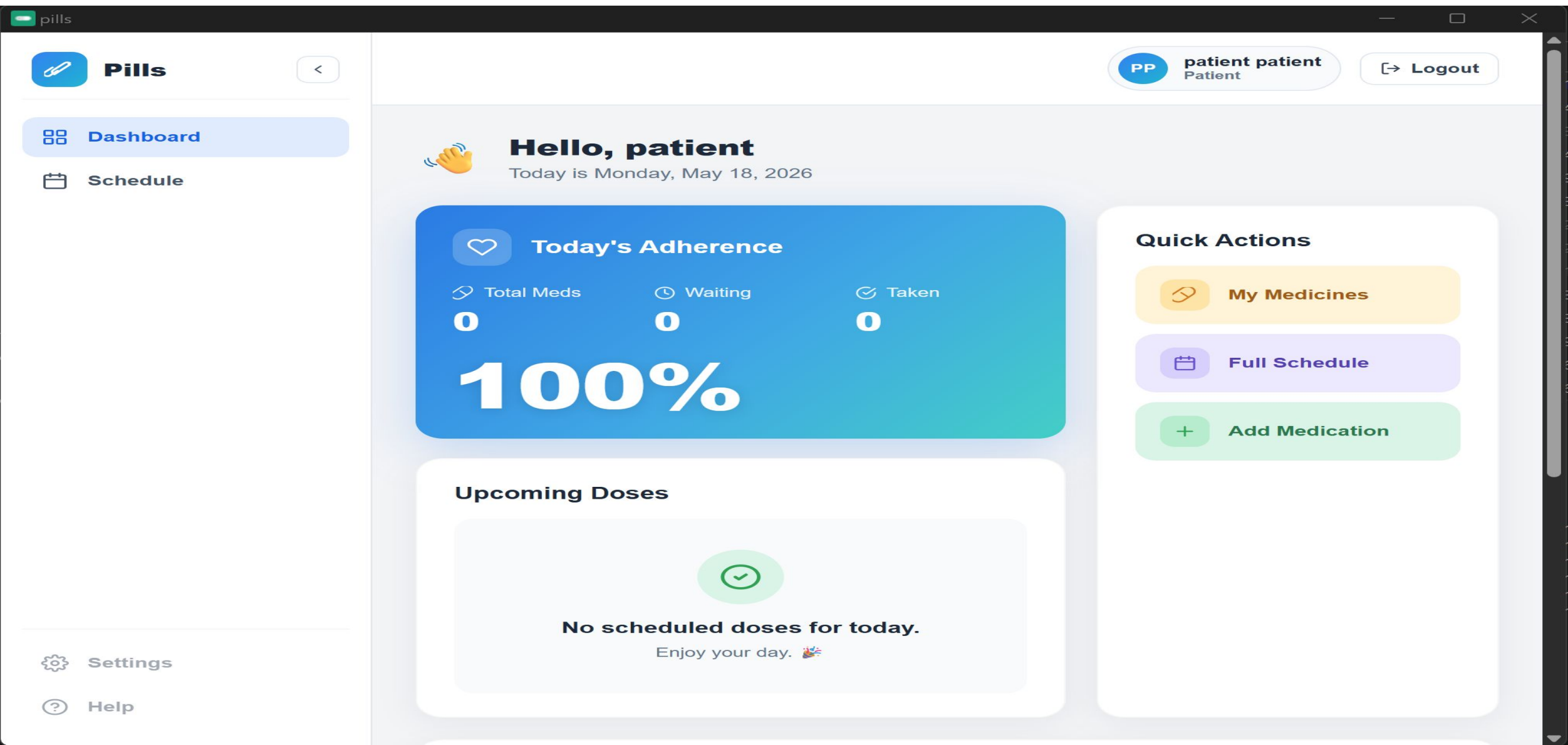
[Forgot password?](#)

Log In

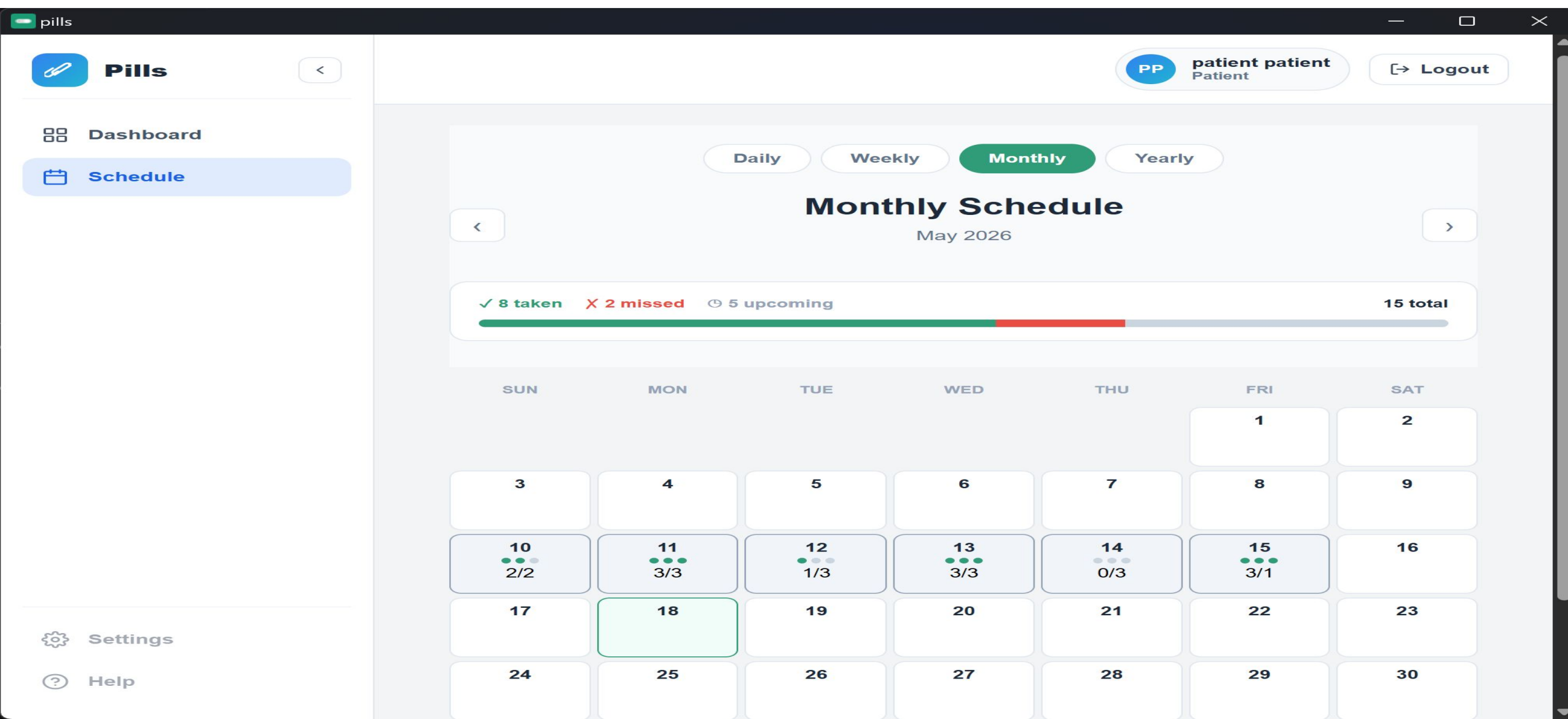
Don't have an account?

[Create New Account](#)

UI Design: Patient Dashboard



UI Design: Schedule & Calendar



Caregiver UI: Adding Medication

pills

Pills

<

Dashboard

Patients

Schedules

Edit Schedules

Users

Settings

Help

CCcare careCaregiver

[> Logout]

Medication Schedules

Assign medications to your connected patients

Patient

Select a patient

Medication Name

Type at least 3 letters...

Medication Type

Pill

Dosage

e.g. 500mg

Quantity

1

Every (hours)

8

Start hour

08:00 AM

Start date

mm/dd/yyyy

End date

mm/dd/yyyy

Instructions

e.g. after meal, with water

Add Schedule

Caregiver UI: Manage Schedules

 Pills

<

 Dashboard

 Patients

 Schedules

 Edit Schedules

 Users

 Settings

 Help

CCcare careCaregiver

[>] Logout

Edit Schedules

View and modify medication schedules for your patients

Patient

patient patient (p@p.com)

PANTOMAX 40MG E.C.TABLETS (Other)

Dosage: - - | Quantity: 1

4/10/2026 – 5/1/2026

03:01 PM · every 8 hour(s)

Instructions: water

Edit

Panadol (Pill)

Dosage: 500 mg | Quantity: 1

4/21/2026 – 4/25/2026

11:06 AM · every 8 hour(s)

Instructions: after food

Edit

PANTOMAX 40MG E.C.TABLETS (Other)

Technology Stack: Backend (Go)

Go (Golang) was selected for the backend due to its exceptional reliability and performance:



Built-in Concurrency

Utilizes goroutines to efficiently manage background notification schedulers without UI lag.



Fast Compilation

Compiles rapidly to native machine code, ensuring maximum performance on older desktop hardware.



Cross-Platform Compilation

Allows building native executables for Windows, macOS, and Linux from a single machine.



Security

Strong static typing catches errors at compile-time; handles SQLite encryption seamlessly.

Technology Stack: Framework (Wails)

Wails v2.11 bridges the Go backend with web technologies, providing a superior alternative to Electron:



Native Desktop Apps

Produces truly native apps using the OS's native webview, not a bundled Chromium browser.



Seamless Integration

Go functions are automatically converted to typed JavaScript functions for immediate frontend use.



Extremely Small Binaries

Results in drastically smaller file sizes (e.g., 4MB on macOS vs 120MB+ on Electron).



Fast Startup

Enables the application to easily launch in under 3 seconds.

Technology Stack: Frontend (Svelte)

Svelte was selected over React/Vue due to compile-time optimizations and accessibility features:



Reactive by Default

UI updates happen in real-time (like the adherence percentage bar) instantly.



No Virtual DOM

Compiles declarative components directly into highly efficient JavaScript that manipulates the DOM.



Small Footprint

Contributes to reducing the overall size of the final executable file.



Built-in Accessibility

The Svelte compiler issues warnings for poor accessibility practices during development.

Development Workflow: Repository

 nashamri / pills



+













<> Code

 Issues 5

 Pull requests

 Agents

 Actions

 Projects

 Security and quality

 Insights

 Settings

 pills

Private

 Watch 0

 Fork 0

 Star 0

 main

 2 Branches

 4 Tags

T

Add file

<> Code

 nashamri UI tweaks	4b2d40c · 16 hours ago	 97 Commits
 .github/workflows	github action: embed webview2 for windows	3 days ago
 pills	UI tweaks	16 hours ago
 .gitignore	add wails	4 months ago
 shell.nix	add action to build on all 3 platforms	3 days ago

 README



Add a README

About

No description, website, or topics provided.

-  Activity
-  0 stars
-  0 watching
-  0 forks

Releases 3

 v0.0.3

Latest

3 days ago

+ 2 releases

Packages

No packages published

Development Workflow: Reviews & CI

nashamri / pills

🔍 Type / to search

<> Code

⌚ Issues 5

🔗 Pull requests

👤 Agents

▶ Actions

📅 Projects

🛡 Security and quality

📊 Insights

⚙ Settings

Filters 🔻

🔍 is:pr is:closed

🏷 Labels 9

📌 Milestones 0

New pull request

✕ Clear current search query, filters, and sorts

☐

🔗 0 Open

✓ 34 Closed

Author 🔻

Label 🔻

Projects 🔻

Milestones 🔻

Reviews 🔻

Assignee 🔻

Sort 🔻

☐

🔗

fix release action ✓

#64 by nashamri was merged 3 days ago

☐

🔗

Improvements to calendar display ✓

#63 by 404a70x was merged 3 days ago

☐

🔗

تم اضافة خيار تعديل الدواء لمقدم الرعاية ✓

#62 by Saud-ALdandn was merged 3 days ago

☐

🔗

User-ui1

#59 by fahadalshammari1 was closed last week

💬 1

☐

🔗

عرض ملخص الجرعات في التقويم الشهري بشكل صحيح ✓

#58 by 404a70x was merged last week

💬 3

☐

🔗

Admin تظهر فقط ل View User جعل واجهة ✓

#57 by Saud-ALdandn was merged last week

☐

🔗

تم اضافة اسم المستخدم ودوره في الواجهة ✓

#56 by fahadalshammari1 was merged 5 days ago

☐

🔗

اضافة بعض المميزات على صفحة الجدول الزمني ✓

#50 by 404a70x was merged 3 weeks ago

☐

🔗

pages الى مجلد calendar تم نقل ملف ✓

Conclusion and Future Work

We have successfully delivered a reliable, accessible, and free solution that bridges the gap between elderly patients and caregivers. Local SQLite storage and the Wails/Svelte architecture have proven highly effective for secure healthcare tools.

Future Roadmap:



System Notifications Integration:

Enable native system alerts even when the application is closed.



Full Arabic Support (RTL):

Implement translation indices for full UI simulation from right to left.



Drug Interaction API:

Integrate external databases to warn caregivers about dangerous drug combinations.

Questions?

Thank you for your attention.

Code Repository: github.com/nashamri/pills

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